

# AI in Production — Building & Scaling Real-World Intelligent Software Systems

Theme: Build, deploy, and scale intelligent software solutions that integrate Artificial Intelligence into real-world applications — taking AI models from development to production through robust software architectures, APIs, cloud platforms, and deployment pipelines.

DOMAIN

AI Engineering & MLOps

CATEGORY

SOFTWARE

PS TYPE

Open Innovation

## BACKGROUND

A large share of AI projects today never move beyond a notebook or a demo. Teams can train an impressive model, yet struggle to turn it into something reliable that real users can depend on — one that handles live traffic, scales under load, stays available, and keeps working as data and usage patterns change. The gap between "a model that works once" and "a system that works every time, for everyone" is where most real-world AI initiatives quietly fail.

## PROBLEM STATEMENT

Design and build an end-to-end software solution that takes an AI/ML capability from a working model to a production-grade application — one that is exposed through clean APIs, deployed on a cloud platform, and engineered to be reliable, observable, and scalable enough for real users and real workloads, rather than a one-off script or a single demo run.

## EXPECTED FOCUS AREAS

- Wrapping AI/ML models behind well-designed, versioned APIs and services
- CI/CD pipelines for continuous model and application delivery
- Handling latency, concurrency, and cost trade-offs at scale
- Containerized, cloud-native deployment with autoscaling and load balancing
- Monitoring, logging, and alerting for model performance and system health
- Security, versioning, and graceful fallback when a model underperforms or fails

## EXPECTED OUTCOME

A working, deployed application that demonstrates a real AI capability (such as prediction, recommendation, classification, or generation) integrated into a usable product — accessible via an API or interface, running on cloud infrastructure, and able to handle realistic traffic without manual intervention.

Solutions will be evaluated on architectural soundness, reliability under load, ease of integration, observability, and how convincingly the system reflects real production practice rather than a one-time demo.

## IMPACT

### RELIABILITY

Turns experimental AI models into dependable software that real users and businesses can trust.

### SCALABILITY

Enables AI-powered products to serve many users concurrently without breaking down.

### REAL-WORLD READINESS

Bridges the gap between research prototypes and deployable, production-grade AI applications.